

Python Examples — Jupyter Notebook

In [1]:

```
# Print Hello Name
name = input("Enter your name: ")
print("Hello", name)
```

Out [1]:

```
Enter your name: Tanvi
Hello Tanvi
```

In [2]:

```
# Addition
a = int(input("Enter value of a: "))
b = int(input("Enter value of b: "))
print("Sum =", a + b)
```

Out [2]:

```
Enter value of a: 4
Enter value of b: 7
Sum = 11
```

In [3]:

```
# Subtraction
a = int(input("Enter value of a: "))
b = int(input("Enter value of b: "))
print("Subtraction =", a - b)
```

Out [3]:

```
Enter value of a: 9
Enter value of b: 1
Subtraction = 8
```

In [4]:

```
#Multiplication
a = int(input("Enter value of a: "))
b = int(input("Enter value of b: "))
print("Multiplication =", a * b)
```

Out [4]:

```
Enter value of a: 6
Enter value of b: 8
Multiplication = 48
```

In [5]:

```
# Division
a = float(input("Enter value of a: "))
b = float(input("Enter value of b: "))
n = int(input("Enter decimal places: "))
```

```
if b != 0:
    print("Division =", round(a / b, n))
else:
    print("Division by zero not possible")
```

Out [5]:

```
Enter value of a: 78
Enter value of b: 6
Enter decimal places: 3
Division = 13.0
```

In [6]:

```
# Swap Two Variables
a = int(input("Enter value of a: "))
b = int(input("Enter value of b: "))
print("Before Swapping:")
print("a =", a)
print("b =", b)
a, b = b, a
print("After Swapping:")
print("a =", a)
print("b =", b)
```

Out [6]:

```
Enter value of a: 34
Enter value of b: 2
Before Swapping:
a = 34
b = 2
After Swapping:
a = 2
b = 34
```

In [7]:

```
# Check Even or Odd
n = int(input("Enter value of n: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Out [7]:

```
Enter value of n: 576
Even
```

In [8]:

```
# Positive or Negative number
n = int(input("Enter value of n: "))
if n > 0:
    print("Positive")
```

```
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Out [8]:

```
Enter value of n: -192
Negative
```

In [9]:

```
# Reverse Number for Negative Values
n = int(input("Enter number: "))

sign = -1 if n < 0 else 1
n = abs(n)

reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10

print("Reverse =", reverse * sign)
```

Out [9]:

```
Enter number: 12345
Reverse = 54321
```

In [13]:

```
# Count Digits in a Number
n = int(input("Enter number: "))
print("Total Digits =", len(str(abs(n))))
```

Out [13]:

```
Enter number: 178912782
Total Digits = 9
```

In [18]:

```
# print number 1 to n column format
n = int(input("Enter value of n: "))
for i in range(1, 11):
    for j in range(i, n + 1, 10):
        print(j, end="\t")
    print()
```

Out [18]:

```
Enter value of n: 100
1 11 21 31 41 51 61 71 81 91
2 12 22 32 42 52 62 72 82 92
```

```
3 13 23 33 43 53 63 73 83 93
4 14 24 34 44 54 64 74 84 94
5 15 25 35 45 55 65 75 85 95
6 16 26 36 46 56 66 76 86 96
7 17 27 37 47 57 67 77 87 97
8 18 28 38 48 58 68 78 88 98
9 19 29 39 49 59 69 79 89 99
10 20 30 40 50 60 70 80 90 100
```

In [19]:

```
# print Table upto 1 to n in column format
n = int(input("Enter last table number: "))
for i in range(1, 11):
    for j in range(1, n + 1):
        print(f"{j} x {i} = {j*i:<4}", end="\t")
    print()
```

Out [19]:

```
Enter last table number: 5
1 x 1 = 1 2 x 1 = 2 3 x 1 = 3 4 x 1 = 4 5 x 1 = 5
1 x 2 = 2 2 x 2 = 4 3 x 2 = 6 4 x 2 = 8 5 x 2 = 10
1 x 3 = 3 2 x 3 = 6 3 x 3 = 9 4 x 3 = 12 5 x 3 = 15
1 x 4 = 4 2 x 4 = 8 3 x 4 = 12 4 x 4 = 16 5 x 4 = 20
1 x 5 = 5 2 x 5 = 10 3 x 5 = 15 4 x 5 = 20 5 x 5 = 25
1 x 6 = 6 2 x 6 = 12 3 x 6 = 18 4 x 6 = 24 5 x 6 = 30
1 x 7 = 7 2 x 7 = 14 3 x 7 = 21 4 x 7 = 28 5 x 7 = 35
1 x 8 = 8 2 x 8 = 16 3 x 8 = 24 4 x 8 = 32 5 x 8 = 40
1 x 9 = 9 2 x 9 = 18 3 x 9 = 27 4 x 9 = 36 5 x 9 = 45
1 x 10 = 10 2 x 10 = 20 3 x 10 = 30 4 x 10 = 40 5 x 10 = 50
```

In [20]:

```
# Sum of First n Numbers
n = int(input("Enter value of n: "))
total = 0
for i in range(1, n + 1):
    total += i
print("Sum =", total)
```

Out [20]:

```
Enter value of n: 10
Sum = 55
```

In [22]:

```
# Basic factorial program
n = int(input("Enter value of n: "))
if n < 0:
    print("Factorial not possible")
else:
    fact = 1
    for i in range(1, n + 1):
```

```
fact *= i
print("Factorial =", fact)
```

Out [22]:

```
Enter value of n: 6
Factorial = 720
```

In [24]:

```
# Fibonacci Series
n = int(input("How many terms: "))
a = 0
b = 1
for i in range(n):
    print(a)
    a, b = b, a + b
```

Out [24]:

```
How many terms: 6
0
1
1
2
3
5
```

In [26]:

```
#Palindrome Number
n = input("Enter a number: ")

if n == n[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Out [26]:

```
Enter a number: 1234321
Palindrome
```

In [28]:

```
#Prime Number Check
n = int(input("Enter number: "))
if n > 1:
    for i in range(2, int(n**0.5) + 1):

        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
else:
```

```
print("Not Prime")
```

Out [28]:

```
Enter number: 71
Prime
```

In [30]:

```
# Print Prime Numbers from 1 to n
n = int(input("Enter value of n: "))
for num in range(2, n + 1):
    is_prime = True
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(num)
```

Out [30]:

```
Enter value of n: 25
2
3
5
7
11
13
17
19
23
```

In [31]:

```
# Power Program
a = int(input("Enter base number: "))
b = int(input("Enter power value: "))
result = a ** b
print(f"{a} raised to power {b} = {result}")
```

Out [31]:

```
Enter base number: 4
Enter power value: 2
4 raised to power 2 = 16
```

In [33]:

```
# Reverse a String
text = input("Enter a string: ")
print("Reverse =", text[::-1])
```

Out [33]:

```
Enter a string: board
Reverse = draob
```

In [34]:

```
# Count Vowels in String
text = input("Enter a string: ")
count = 0
for ch in text.lower():
    if ch in "aeiou":
        count += 1
print("Total Vowels =", count)
```

Out [34]:

```
Enter a string: alphabets
Total Vowels = 3
```

In [35]:

```
# Count Alphabets in String Except Spaces and special characters
text = input("Enter a string: ")
count = 0
for ch in text:
    if ch != " " and ch.isalpha():
        count += 1
print("Total Alphabets =", count)
```

Out [35]:

```
Enter a string: this is a flower
Total Alphabets = 13
```

In [36]:

```
# Convert String to Uppercase
text = input("Enter a string: ")
print(text.upper())
```

Out [36]:

```
Enter a string: my name is ram
MY NAME IS RAM
```

In [37]:

```
# Convert String to Lowercase
text = input("Enter a string: ")
print(text.lower())
```

Out [37]:

```
Enter a string: MY NAME IS RAM
my name is ram
```

In [38]:

```
# Palindrome String
text = input("Enter a string: ")
if text == text[::-1]:
    print("Palindrome")
else:
```

```
print("Not Palindrome")
```

Out [38]:

```
Enter a string: kanak
Palindrome
```

In [43]:

```
# Count Words in String
text = input("Enter a sentence: ")
words = text.split()
print("Total Words =", len(words))
```

Out [43]:

```
Enter a sentence: hello tanvi
Total Words = 2
```

In [40]:

```
# Remove Spaces from String
text = input("Enter a string: ")
print(text.replace(" ", ""))
```

Out [40]:

```
Enter a string: she is a pretty women
sheisaprettywomen
```

In [42]:

```
# Find Length of String
text = input("Enter a string: ")
print("Length =", len(text))
```

Out [42]:

```
Enter a string: hello tanvi
Length = 11
```

In [44]:

```
# First Character Capital
text = input("Enter a string: ")
print(text.capitalize())
```

Out [44]:

```
Enter a string: tanvi
Tanvi
```

In [45]:

```
# Sort Characters of String
text = input("Enter string: ")
print("".join(sorted(text)))
```

Out [45]:

```
Enter string: hello tanvi
aehillnotv
```


In [46]:

```
# Sort Words by First Letter
text = input("Enter a sentence: ")
words = text.split()
words.sort()
print("Sorted Words =", words)
```

Out [46]:

```
Enter a sentence: hello tanvi
Sorted Words = ['hello', 'tanvi']
```

In [47]:

```
# Find Largest Element in List
numbers = list(map(int, input("Enter numbers separated by space: ").split()))
largest = max(numbers)
print("Largest Element =", largest)
```

Out [47]:

```
Enter numbers separated by space: 56 34 44 88 100 1 23 42
Largest Element = 100
```

In [48]:

```
# Find Smallest Element in List
numbers = list(map(int, input("Enter numbers separated by space: ").split()))
smallest = min(numbers)
print("Smallest Element =", smallest)
```

Out [48]:

```
Enter numbers separated by space: 56 34 44 88 100 1 23 42
Smallest Element = 1
```

In [50]:

```
# Create Dictionary from User Input
student = {}
n = int(input("How many entries: "))
for i in range(n):
    key = input("Enter key: ")
    value = input("Enter value: ")
    student[key] = value
print(student)
```

Out [50]:

```
How many entries: 2
Enter key: name
Enter value: tanvi
Enter key: age
Enter value: 10
{'name': 'tanvi', 'age': '10'}
```

In [52]:

```
# Access Dictionary Value
```

```
student = {  
    "name": "Komal",  
    "age": 21,  
    "city": "Bikaner"  
}  
key = input("Enter key to search: ")  
print(student.get(key, "Key Not Found"))
```

Out [52]:

```
Enter key to search: class  
Key Not Found
```

In [54]:

```
# Remove Duplicate Values While Preserving Order  
numbers = list(map(int, input("Enter numbers: ").split()))  
unique = []  
for num in numbers:  
    if num not in unique:  
        unique.append(num)  
print(unique)
```

Out [54]:

```
Enter numbers: 1 2 3 5 7 1 7 9 10  
[1, 2, 3, 5, 7, 9, 10]
```

In [57]:

```
# Check Leap Year  
year = int(input("Enter year: "))  
  
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):  
    print("Leap Year")  
else:  
    print("Not Leap Year")
```

Out [57]:

```
Enter year: 2024  
Leap Year
```

In [62]:

```
# Armstrong Number: Digits ki power ka total = original number  
#153 = 13+53+33 = 1+ 125+ 27  
n = int(input("Enter number: "))  
temp = n  
power = len(str(n))  
total = 0  
while temp > 0:  
    digit = temp % 10  
    total += digit ** power  
    temp //= 10
```

```
temp //= 10
if total == n:
    print("Armstrong Number")
else:
    print("Not Armstrong Number")
```

Out [62]:

```
Enter number: 153
Armstrong Number
```

In [63]:

```
#Celsius to Fahrenheit
celsius = float(input("Enter Celsius: "))
fahrenheit = (celsius * 9/5) + 32
print("Fahrenheit =", fahrenheit)
```

Out [63]:

```
Enter Celsius: 27
Fahrenheit = 80.6
```

In [68]:

```
# Greatest Among Three Numbers
a = int(input("Enter a: "))
b = int(input("Enter b: "))
c = int(input("Enter c: "))
print("Greatest =", max(a, b, c))
```

Out [68]:

```
Enter a: 44
Enter b: 22
Enter c: 99
Greatest = 99
```

In [69]:

```
# Area of Circle
radius = float(input("Enter radius: "))
area = 3.14 * radius * radius
print("Area =", area)
```

Out [69]:

```
Enter radius: 3
Area = 28.259999999999998
```

In [70]:

```
# Simple Interest
p = float(input("Enter principal: "))
r = float(input("Enter rate: "))
t = float(input("Enter time: "))
si = (p * r * t) / 100
print("Simple Interest =", si)
```

Out [70]:

```
Enter principal: 100
Enter rate: 1
Enter time: 10
Simple Interest = 10.0
```

In [71]:

```
# Find ASCII Value
ch = input("Enter character: ")
print("ASCII Value =", ord(ch))
```

Out [71]:

```
Enter character: A
ASCII Value = 65
```

In [74]:

```
# Remove Vowels from String
text = input("Enter string: ")
result = ""
for ch in text:
    if ch.lower() not in "aeiou":
        result += ch
print(result)
```

Out [74]:

```
Enter string: tajmahal
tjmhl
```

In [75]:

```
# Find Common Elements in Two Lists
a = list(map(int, input("Enter first list: ").split()))
b = list(map(int, input("Enter second list: ").split()))
common = list(set(a) & set(b))
print("Common Elements =", common)
```

Out [75]:

```
Enter first list: 1 4 7 9 10
Enter second list: 2 4 6 8 10
Common Elements = [10, 4]
```

In [77]:

```
# Find Average of 3 Numbers
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
average = (a + b + c) / 3
print("Average =", average)
```

Out [77]:

```
Enter first number: 3
```

```
Enter second number: 6
Enter third number: 9
Average = 6.0
```

In [79]:

```
# Power Table Program (Square / Cube / Any Power)
n = int(input("Enter last number: "))
power = int(input("Enter power value: "))
print(f"\nTable of powers from 1 to {n}\n")
for i in range(1, n + 1):
    print(f"{i}^{power} = {i ** power}")
```

Out [79]:

```
Enter last number: 4
Enter power value: 4

Table of powers from 1 to 4

1^4 = 1
2^4 = 16
3^4 = 81
4^4 = 256
```

In [82]:

```
# Check Voting Eligibility
age = int(input("Enter age: "))
if age >= 18:
    print("Eligible for Voting")
else:
    print("Not Eligible for Voting")
```

Out [82]:

```
Enter age: 17
Not Eligible for Voting
```

In [85]:

```
# Convert Meters into Kilometers
meters = float(input("Enter meters: "))
km = meters / 1000
print("Kilometers =", km)
```

Out [85]:

```
Enter meters: 1789
Kilometers = 1.789
```